

R3 · Isolated 131,072 × 10-bit lookup
1-cycle versus 8-cycle reads · initiation interval: 1 cycle

Platform	Resource	1-cycle read	8-cycle read	Absolute change	Change
Quartus	Estimated ALMs	359	359	+0	+0.00%
Quartus	Registers / FFs	7	84	+77	+1100.00%
Quartus	Inferred RAM bits	1,310,720	1,310,720	+0	+0.00%
Vivado	CLB LUTs	102	111	+9	+8.82%
Vivado	Registers / FFs	3	20	+17	+566.67%
Vivado	BRAM36 equivalents	36	36	+0	+0.00%
Vivado	LUTs used as RAM	0	10	+10	N/A

Synthesis only; this pipeline is not integrated into R1/R2/R4. Unchanged block RAM; no measured Fmax improvement.